Kohn Sham Theory:

KE + Coulombic energy + Exchange correlation functional

(usually what we don't know & we approximate it)

Functional derivative w.r.t density

↓

density to minimize the expression

↓

Kohn-Sham orbitals (approx. to true one-electron orbitals)

" eigenvalues (approximations to the energies of the various one-electron orbitals)

Nørskov d-band Theory (Newns-Anderson Theory)

+ Approximate wavefunction of AB molecule using tight-binding theory

A B

$\Psi_{AB} = c_A\Psi_A + c_B\Psi_B$

where, $\left(-\frac{1}{2}\nabla^2 + v_A\right)\Psi_A = \varepsilon_A\Psi_A$ → one-electron orbital for A

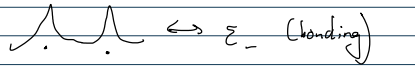
one-electron Hamiltonian: $H = -\frac{1}{2}\nabla^2 + v_A + v_B$

(neglecting $e^- - e^-$ interactions)

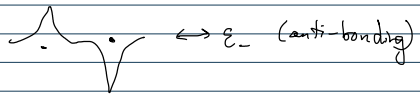
$H\Psi_{AB} = \varepsilon\Psi_{AB}$

↓ Math

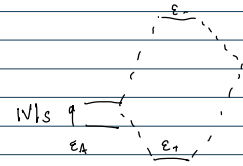
get two solutions:



$\leftrightarrow \varepsilon_+$ (bonding)



$\leftrightarrow \varepsilon_-$ (anti-bonding)



$S = \int \psi_A\psi_B$ } accounts for geom.

$\Delta E \approx 2\varepsilon_+ - \varepsilon_A - \varepsilon_B = -2SV - \sqrt{4V^2 + (\varepsilon_B - \varepsilon_A)^2}$

- $-2SV$: Pauli repulsion (positive)
- $\sqrt{4V^2}$: Covalent contribution to hybridization energy
- $(\varepsilon_B - \varepsilon_A)^2$: Ionic ant to hybridization energy

} Both ionic & covalent attractions matter

Newns-Anderson: Single adsorbate orbital interact w/ single orbital from e^- in solid surface

$$n_a(\epsilon) = \frac{1}{\pi} \frac{\Delta(\epsilon)}{(\epsilon - \epsilon_a - \Lambda(\epsilon))^2 + \Delta(\epsilon)^2}$$ ← result

$\Delta(\epsilon)$ → density of states

- narrow → than adsorbate orbital will be split into bonding & antibonding
- broad → than adsorbate energy level is simply broadened (S & p) (Contribute a constant shift in energy)

Distribution as to why d-orbital has a narrow

↳ S & p has similar energies across all elements → but d bands very a lot and tend to form hybridised orbitals

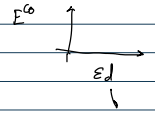
Can integrate DOS to obtain total energy of the system

↳ metal DOS are very narrow (d-bands)

$$\Delta E = -(1-f)\left[\sqrt{4V^2 + (\epsilon_d - \epsilon_f)^2} - (\epsilon_d - \epsilon_f)\right] - 2(1-f)SV$$

f ← occupancy of fermi level

Non-linear curve but has some linearity



d-band center shift

Metal thin-films on substrates:

+Shifts in BE via shifts in d-band centers!

Hammer (1996)
Ruban (1997)

↳ Depositing what on what will